Web Vulnerability Assessment Report

1.Executive Summary

This report presents a detailed analysis of vulnerabilities identified in the DVWA environment. The assessment focused on identifying common web security issues such as SQL Injection and Cross-Site Scripting (XSS). Testing was performed in a controlled lab environment using industry-standard tools.

2. Tools & Environment

- Burp Suite – for intercepting HTTP requests
- DVWA – vulnerable web application for testing
- Web Browser – for interaction
- Kali Linux – testing platform

3. Vulnerability Testing

3.1 SQL Injection

Procedure

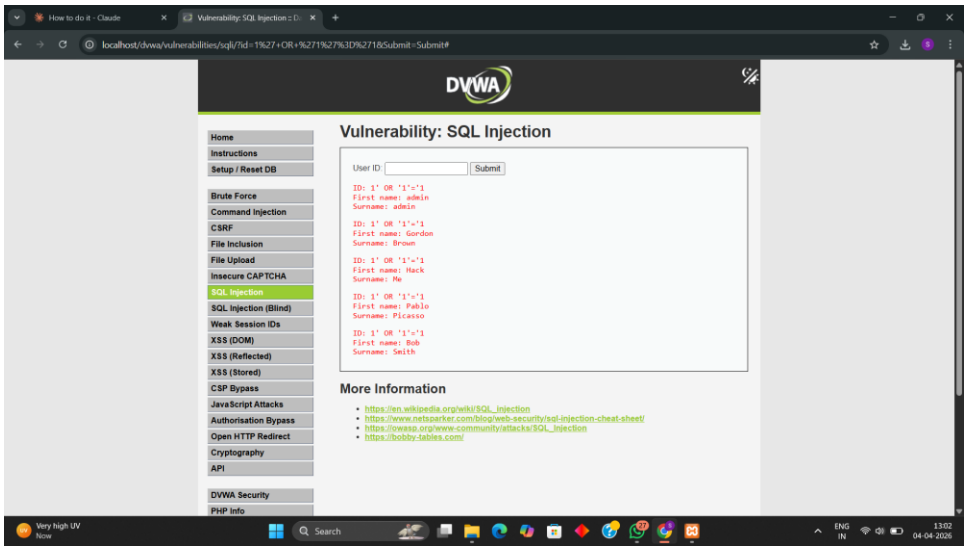
1. Open DVWA and navigate to the SQL Injection module
2. Enter payload: ' OR '1'='1 into the input field
3. Submit the request
4. Observe the application response

Findings

- The application failed to validate user input
- Database query was manipulated successfully
- All records were displayed instead of filtered results
- This confirms vulnerability to SQL Injection

SQL Injection is a critical vulnerability that allows attackers to manipulate database queries. If exploited, it can lead to unauthorized data access, data modification, or complete database compromise.

Result



3.2 Cross-Site Scripting (XSS)

Procedure

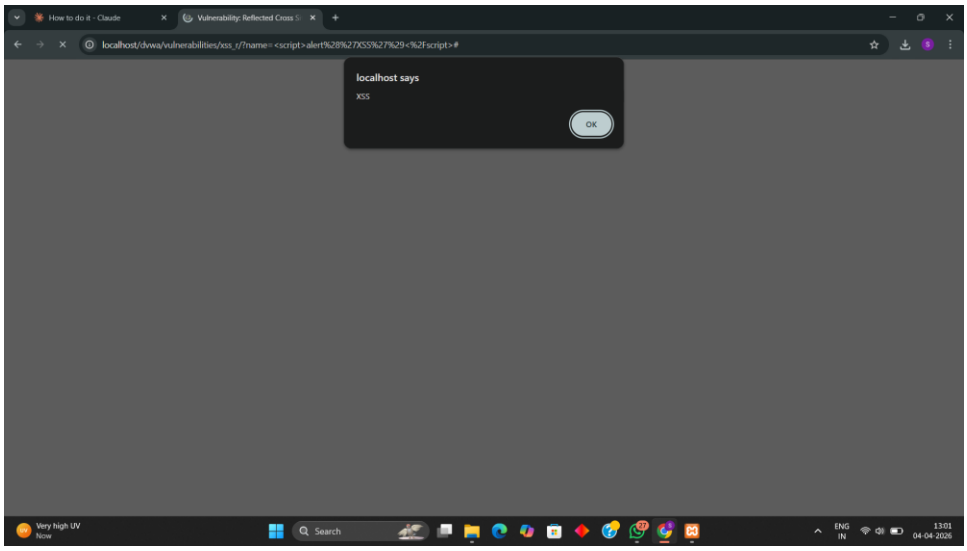
5. Navigate to the XSS module in DVWA
6. Enter payload: <script>alert('XSS')</script>
7. Submit the input
8. Observe the alert popup

Findings

- The application accepted unsanitized input
- JavaScript executed in browser context
- No output encoding was applied
- This confirms XSS vulnerability

Cross-Site Scripting (XSS) allows attackers to inject malicious scripts into web pages viewed by other users. This can result in session hijacking, credential theft, and malicious redirections.

Result



3.3 Burp Suite Interception

Procedure

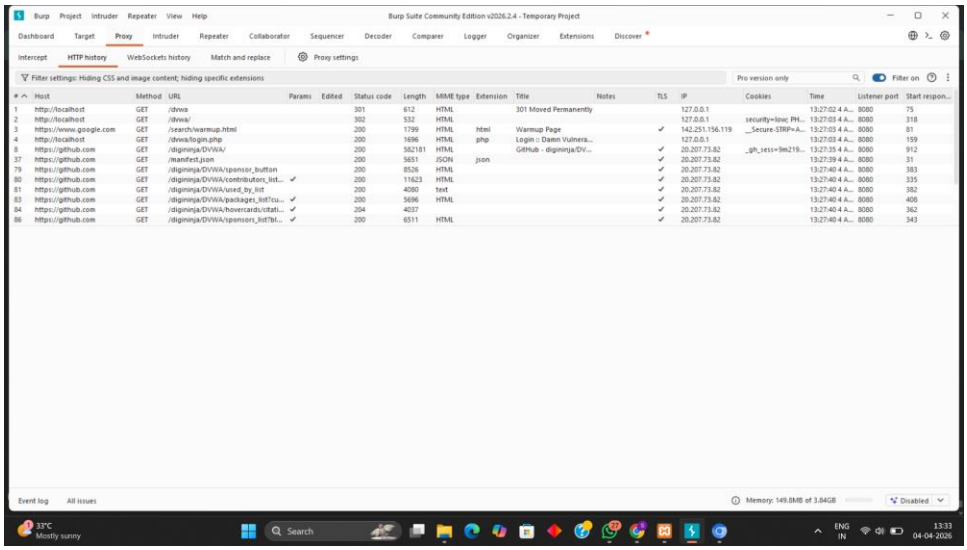
9. Configure browser proxy settings to route traffic through Burp Suite
10. Enable intercept mode in Burp Suite
11. Perform actions in DVWA
12. Capture and analyze HTTP requests

Findings

- HTTP requests and responses were successfully intercepted
- Sensitive parameters were visible in plain text
- Session cookies could be observed
- This indicates risk of session hijacking

Burp Suite demonstrated how attackers can intercept and analyze web traffic. If communication is not properly secured, attackers can exploit this to steal sensitive data or manipulate requests.

Result



4. Conclusion

The assessment identified critical vulnerabilities including SQL Injection and Cross-Site Scripting. These issues arise due to improper input validation and lack of secure coding practices. It is recommended to implement parameterized queries, input sanitization, and secure session handling mechanisms.